

Stat 364 Lab 1

Model Student

Question 2

Create a scatterplot of vehicle weight and horsepower from the `mtcars` data frame. Overlay your plot with a median-median line in red.

```
## this is an example, your code will involve finding the proper
## slope and intercept, not just putting in two numbers
attach(mtcars)
plot(wt, hp, xlab="Weight", ylab="Horsepower")

median_wt<-median(wt)
grp1<-wt<median_wt
table(grp1)
```

```
grp1
FALSE  TRUE
    16    16
```

```
med_wt1<-median(wt[grp1])
med_wt2<-median(wt[!grp1])

med_hp1<-median(hp[grp1])
med_hp2<-median(hp[!grp1])

abline(v=med_wt1, col="blue3", lwd=2)
abline(v=med_wt2, col="blue3", lwd=2)

abline(h=med_hp1, col="orange1", lwd=2)
abline(h=med_hp2, col="orange1", lwd=2)

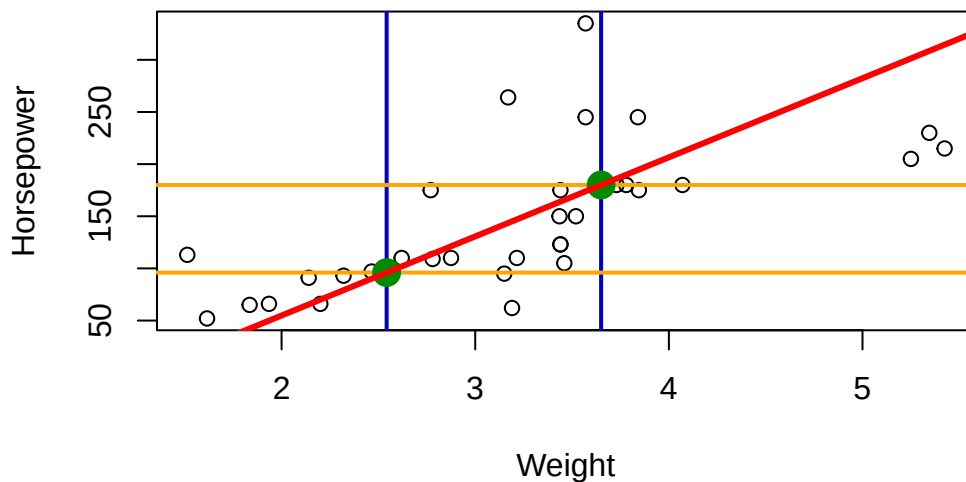
points(c(med_wt1, med_wt2), c(med_hp1, med_hp2), pch=16, cex=2, col="green4")
```

```

rise<-med_hp2-med_hp1
run<-med_wt2-med_wt1
slope<-rise/run

intercept = med_hp1-(slope*med_wt1)
abline(intercept, slope, col="red", lwd=3)

```



Question 3

Create a scatterplot of vehicle speed and stopping distance from the cars data frame. Overlay your plot with a least-squares regression line in the color of your choice.

```
# and now more code here to complete the graphing task
```

Question 5

Fit a least-squares model predicting vehicle hwy_mpg as a function of vehicle weight using the cars04 data frame in the openintro library. Which estimate, d or e , do you feel better about? Why?

And here you type in your answer to the question...